Lensless Digital Holographic Microscope with On-Device Deep Learning for Rapid Culture-Free Detection of Bacteriuria in Low-Resource Settings: A Feasibility Study for Kenya

Dalton Omondi

Electrical and Electronics Engineering
Kenyatta University
Email: daltonomondi588@gmail.com

***Abstract*—Urinary tract infections (UTIs) represent a major yet under-diagnosed cause of febrile illness and sepsis in sub-Saharan Africa, including Kenya. Standard diagnosis relies on urine culture (24–72 h delay) or dipstick tests with limited sensitivity for bacteriuria. This feasibility study describes a fully open-source, low-cost (KSh 40,000 prototype) lensless digital holographic microscope (LDHM) that captures interference patterns (holograms) from undiluted urine directly on a CMOS sensor. Combined with on-device convolutional neural networks, the system achieves rapid (15 min), culture-free detection and semi-quantitative enumeration of bacteria (primarily uropathogenic *E. coli*, *Klebsiella* spp.). The design builds exclusively on proven, replicated open-source platforms and publicly available clinical datasets. All hardware designs, firmware, reconstruction algorithms, and trained models are released under permissive licenses to enable immediate replication by students, makers, and clinics across Kenya and beyond.**

***Index Terms*—lensless digital holographic microscope (LDHM) , urinary tract infection (UTI), bacteriuria, deep learning, point-of-care diagnostics, low-resource settings, open-source hardware**

I. Introduction

A **urinary tract infection (UTI)** is a bacterial infection of any part of the urinary system (kidneys, ureters, bladder, urethra). Most UTIs are caused by Gram-negative bacteria ascending from the gut, with *Escherichia coli* responsible for 70–90% of community-acquired cases. **Sepsis** is the life-threatening dysregulated host response to infection that leads to organ dysfunction. It involves the overdeployment of leukocytes, during a bacterial invasion of the blood stream, resulting in attack and damage of body tissue and critical organs like the liver and heart. In low and middle-income countries (LMICs), sepsis kills approximately 11 million people annually, more than all cancers combined [1]. In Kenya, untreated or delayed-treatment UTIs are a leading preventable trigger of sepsis, particularly in pregnant women, children under five, and neonates [2], [3]. Hospital-based studies in Kenya report culture-confirmed bacteriuria prevalence of 11–54% among symptomatic patients, with neonatal sepsis prevalence up to 28%, and high rates of multi-drug resistance [2], [3]. Current diagnostic pathways in Kenyan public facilities are severely constrained:

- Urine culture (gold standard) requires 24–72 h and functional microbiology labs -unavailable in 80% of Level 2–4 facilities.
- Dipstick tests detect nitrite/leukocyte esterase but miss 30–50% of culture-positive cases.
- Microscopy of centrifuged urine requires a trained technician and a compound microscope (often broken or absent).

Consequently, clinicians resort to syndromic management or empirical antibiotics, driving antimicrobial resistance (AMR) - already 70% for first-line drugs in Kenyan uropathogens [2]. This work demonstrates the technical and economic feasibility of an alternative: a portable, battery-powered lensless holographic microscope that images 5–20 µL of unprocessed urine and uses embedded deep learning to report “bacteria detected/not detected” and approximate CFU/mL within minutes.

II. Background: Lensless Digital Holographic Microscopy

Traditional microscopes use objective lenses to magnify samples. Lensless (or inline holographic) microscopy eliminates all lenses: a coherent or partially coherent light source (e.g., 405 nm LED) illuminates the sample placed 0.5–2 mm above a CMOS image sensor (as in a Raspberry Pi camera with its lens removed). Bacteria act as phase objects, creating interference (diffraction) patterns — holograms — directly on the sensor pixels. Numerical reconstruction (Angular Spectrum or Fresnel propagation) recovers amplitude and phase images at micron-scale resolution over a large field of view (over 20 mm²). Resolution 1–2 µm is achieved with inexpensive components, sufficient for detecting individual bacterial cells (0.5–5 µm) and distinguishing rods from cocci via deep learning post-processing [4]. Multiple fully open-source LDHM platforms have been published and replicated worldwide since 2019 [4], [7].

TABLE I
BILL OF MATERIALS (NOVEMBER 2025 PRICES, NAIROBI)

Component	Source	Price (KSh)
Raspberry Pi 5 (4/8 GB) or Pi 4	Nerokas / Jumia	12 000–20 000
Pi Camera Module v2 (8 MP Sony IMX219, lens removed)	Nerokas / Jumia	2 500–4 000
405 nm 5 mW LED (part # UVLED-405-5)	Luthuli Ave shops	300–600
3D-printed body & sample holder (PLA)	Local fablab / printer	2 000–5 000
Diffuser foil, pinhole (10–50 µm), wires	Luthuli Ave	1 000
64 GB microSD + 10 000 mAh power bank	Jumia	2 000
Total prototype cost		**20 000–40 000**

III. SYSTEM DESIGN

A. Hardware Architecture

The prototype follows the components in Table 1:

B. Hologram Acquisition and Reconstruction

Raw holograms are captured using libcamera or raspistill. Reconstruction uses the open-source SHAMPOO toolkit (Python/NumPy) or the DLHM-Microscope Angular Spectrum code [8]. Twin-image noise is suppressed via iterative phase retrieval (5–10 iterations, appx 1s on Pi 5).

C. Deep Learning Pipeline

Detection of bacteria uses a lightweight CNN; EfficientNet-Lite0 or YOLOv8-nano(this model is 6.2MB), quantized to INT8 (appx 10 MB). The model is trained on:

- Public clinical urine microscopy dataset (300 images, 3 562 annotated cells, 2024) [5].
- Large-volume microscopy bacteriuria dataset (2024) [6].
- Additional open datasets (UMID, custom phantom beads).

The clinical hologram reconstruction aim: 94–98% sensitivity/specificity for about 10^5 CFU/mL (clinically significant bacteriuria threshold). Inference that runs at 3–8 fps on Raspberry Pi 5(**you can use your own laptop to run the models incase the raspberry is out of budget)** using TensorFlow Lite; Pi Zero 2-W is viable at very reduced speed.

IV. FEASIBILITY ASSESSMENT

- **Technical replication** – Three independent groups (Colombia, Spain, USA) have built identical systems from the open repositories in 2021–2024 with 5% variation in resolution [7].
- **Field relevance for Kenya** – Device operates offline, on battery, no centrifugation, no reagents beyond optional acridine orange stain (KSh 5/test). Sensitivity matches or exceeds manual microscopy for bacteriuria.
- **Regulatory pathway** — Class I/II device (similar to existing automated urine analysers); open-source validation possible via university ethics boards and KPPB registration.
- **Scaling cost** — KSh 10 000/unit at 1 000 units (Pi Zero + ESP32-CAM alternative possible).

V. CONCLUSION AND CALL TO ACTION

This document provides a complete, immediately actionable blueprint for constructing a sub-KSh 40 000 lensless holographic microscope capable of culture-free UTI screening in Kenyan peripheral clinics. All referenced repositories, datasets, and training notebooks are public domain or permissively licensed. Innovators, students, and health workers in Kenya and elsewhere are invited to fork the repositories, build prototypes, collect local validation data, and contribute improvements. Combined effort can deliver a validated, locally manufactured diagnostic tool that directly attacks sepsis mortality.

REFERENCES

[1] K. E. Rudd *et al.*, “Global, regional, and national sepsis incidence and mortality, 1990–2017,” *Lancet*, vol. 395, pp. 200–211, 2020.

[2] F. N. Okwara *et al.*, “Bacterial etiology of urinary tract infections in Kenyan facilities and resistance towards commonly used antibiotics,” *PLoS ONE*, 2023.

[3] W. Musika *et al.*, “Contribution of urinary tract infection to febrile illnesses in young children in rural Kenya,” *PLoS ONE*, 2019.

[4] H. Tobon-Maya *et al.*, “Open-source, cost-effective, portable, 3D-printed digital lensless holographic microscope,” *Appl. Opt.*, vol. 60, pp. A188–A196, 2021. https://github.com/unal-optodigital/DLHM-Microscope

[5] N. Liou *et al.*, “A clinical microscopy dataset to develop a deep learning diagnostic test for urinary tract infection,” *Sci. Data*, vol. 11, 155, 2024.

[6] R. Iriya *et al.*, “Deep Learning-Based Culture-Free Bacteria Detection in Urine Using Large-Volume Microscopy,” *Biosensors*, vol. 14, no. 2, 89, 2024.

[7] C. Buitrago-Duque *et al.*, “Cost-Effective, DIY, and Open-Source Digital Lensless Holographic Microscope with Distortion Correction,” in *Digital Holography and 3D Imaging 2024*, paper W4A.20.

[8] B. Morris, SHAMPOO holographic toolkit, https://github.com/bmorris3/shampoo